Name: _____ Date: _____

Math 2710 — Worksheet

2026-09-24

Show your work. Write your final answer clearly.

1. Decide whether or not the following pair of statements are logically equivalent. Justify your answer.

$$\sim (P \Rightarrow Q) \qquad \text{and} \qquad P \wedge \sim Q.$$

2. Write the following statement as an English sentence, and say whether it is true or false.

$$\exists a \in \mathbb{R},\ \forall x \in \mathbb{R},\ ax = x.$$

3. Write the following statement as an English sentence, and say whether it is true or false.

$$\forall n \in \mathbb{Z},\ \exists X \subseteq \mathbb{N},\ |X| = n.$$